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Rainfall Monitoring Using Cellular Networks

Recently, several media articles discussed the work of researchers in Israel who used wireless networks to estimate rainfall patterns [1]. Their method uses changes in the electromagnetic field measured by existing mobile phone networks; therefore, it does not involve additional costs. In what follows, we summarize the applications of the rainfall pattern estimation, the signal processing characteristics and challenges of the new environmental monitoring tool, and potential improvements.

The estimation of spatial-temporal rainfall has important implications in meteorology, hydrology, agriculture, environmental policies, and weather forecasting. Typically, rainfall is monitored using a network of rain gauges and weather radars. Rain gauges provide accurate ground-level fixed-point location measurements. However, they do not provide sufficient spatial resolution data, and the measurements are prone to errors caused by wind, humans, and other factors. On the other hand, weather radars estimate rainfall field over large contiguous areas, provide high-resolution temporal data, and wide coverage of spatial data, but are not as accurate as rain gauges due to factors such as differences between measuring aloft and at ground level, nonweather echoes, variations of reflectivity with height, and clutters.

The idea of using measurements of microwave electromagnetic field attenuations for rainfall estimation based on specially designed equipment has been explored in the past. A tomographic technique that uses received signal level (RSL) measurements from a set of microwave links with predefined geometry, transmitting on a specially selected frequency, was proposed in [2]. It is known that the microwave radio propagation is largely influenced by scattering and absorption due to water droplets, especially at frequencies above 10 GHz where the radio wavelengths are close to the raindrop sizes. In wireless communication, these propagation impairments are portrayed as large variations in the received signal level. To provide reliable communication links in spite of these impairments, researchers have invested much effort in developing models and mechanisms using long-term precipitation statistics, while modem companies have developed complicated fade mitigation techniques. The "Microwave Attenuation as a New Tool for Improving Storm-Water Supervision Administration" (MANTISSA) project, which involves several European universities, has explored the use of dual-frequency microwave links to estimate rainfall [3]. However, the use of existing wireless communication systems for rainfall monitoring brings opportunities as well as demanding challenges; the frequency of the signals, as well as the geometry of the links, their lengths, and the RSL measurement protocols, were designed to optimize the communication tasks and cannot be optimized for environmental monitoring purposes.

In existing wireless transmission networks that comprise fixed line-of-sight microwave links (backhaul links), network operators keep track of the RSL for failure monitoring or fade mitigation purposes. Timely attenuation data can be recorded from every link in the monitored area and transmitted to a central processing facility. Due to the fact that the attenuation data is monitored for failure purposes, the RSL quantization resolution is coarse (1 dB) and the time resolution is typically a system parameter defined by an equipment manufacturer.

Signal processing methods are applied to analyze the collected measurements, reduce scintillation effects by appropriate prefiltering, and find clear-sky attenuation so that each link power measurement is converted to rain rate using a model. The commonly used model for the rain-induced attenuation for a terrestrial line-of-sight (LoS) microwave integrated path is given by $A = aR^b$, where A is the attenuation expressed in dB/km, R denotes the rain rate along the link measured in mm/h, and the constants a and b are (in general) functions of the precipitation type, frequency, polarization, and temperature. The validity of the model is now fairly well established. Although the model is regarded as empirical, a strong theoretical justification exists for the choice of these parameters [4], [5].

Other challenging signal processing tasks arise from the fact that the geometry and other parameters of the monitoring system, designed by cellular operators to maximize communication performance, are arbitrary. Implementing a technique capable of processing multiple path-integrated rain rates to derive an estimate of the two-dimensional rainfall intensity field is not straightforward: the model has to deal with the large quantization errors as well as with different links lengths, frequencies, and polarizations, unknown drop size distribution, and arbitrary geometry of the network.

To reconstruct the spatial-temporal precipitation distribution using cellular backhaul data, a tomography-based method is employed. As illustrated in Figure 1(b), a rainfall intensity map

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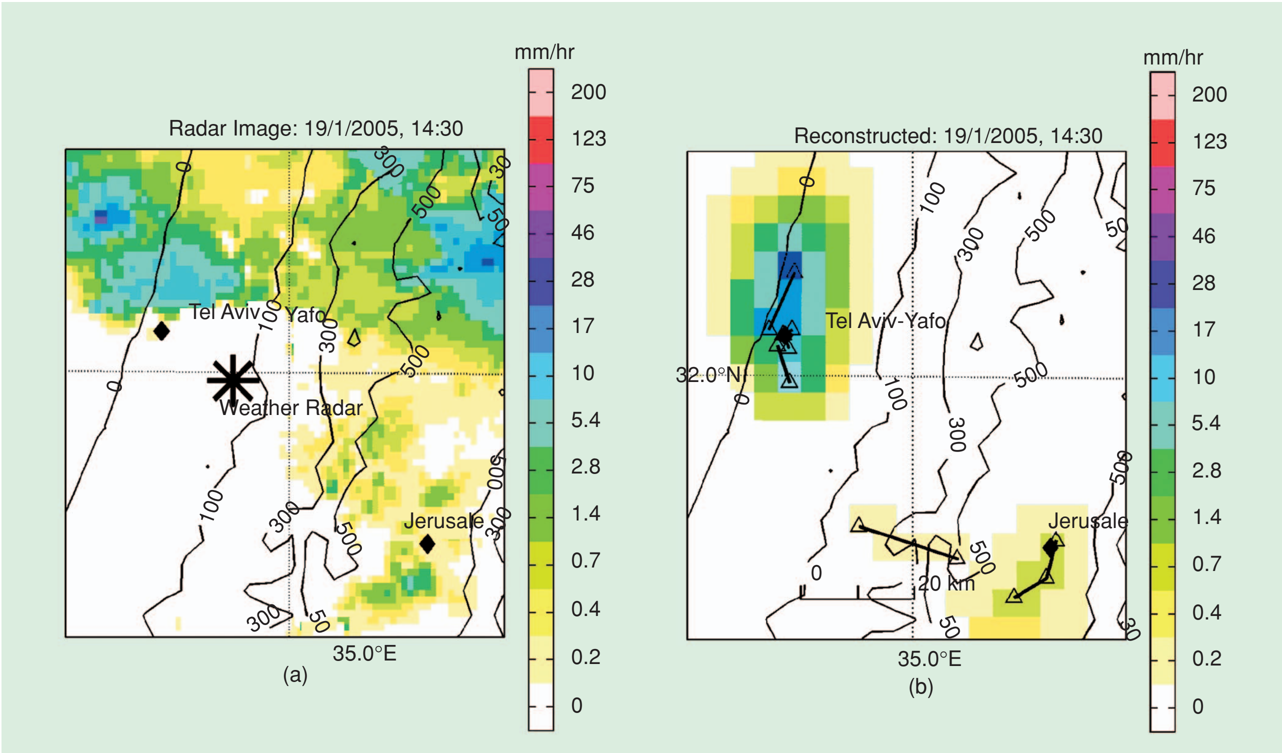

[FIG1] Maps of rainfall intensities over central Israel on 19 January 2005: (a) from weather radar images and (b) calculated using RSL measurements from seven cellular backhauls. The two maps cover the same area at the same time slot.

estimate is produced using seven cellular links. The link density varies from 0.3 and less, up to 3 links/km^2. The cellular backhauls rainfall estimation results, which are restricted to areas where measurements/links are available, are more accurate in the sense that they correlate better with the "ground truth" rain gauges results [1] than the weather radar results in Figure 1(a). On the other hand, in this example, where only seven backhaul links have been used, the weather radar results in Figure 1(a) demonstrate higher spatial resolution, although the radar map does not cover the whole area due to unavoidable clutter. The radar clutter is due to topographical barriers (such as a mountain system west of Jerusalem) or human barriers (for instance, urban regions).

To address the spatial resolution limitation of the cellular backhauls, stochastic interpolation techniques that can be integrated with rain gauges and weather radar-based methods are being developed. Other current and future challenges include: a) the usage of measurements from power control protocols with the mobile phones (handsets rather than backhaul transmission) for precipitation monitoring and b) the integration of accurate high-resolution cellular precipitation measurements into mathematical models used for weather forecasting. In a), the difficulties arise from 1) cellular radio links being weakly sensitive to the rainfall at mobile operation frequencies (1–2 GHz), 2) lack of LoS (shadowing), and 3) fast fading due to terminal movement dominating the attenuation due to changing environmental conditions. In b), the goal is to improve the model initialization. While the equations that describe the change of the atmospheric state (three-dimensional distributions of temperature, moisture, density, pressure, and wind) are relatively well understood, the initialization of the model with poor and incomplete data results in inaccurate prediction. Assimilation of cellular precipitation dynamic measurements using Kalman filtering can help reduce uncertainties in estimation of the true state of the atmosphere and, eventually, improve forecasting of floods and storms.

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